

# Software Design Document - Healthcare Appointment System

## Project Overview

A healthcare portal enabling patient registration, appointment booking, doctor schedules, prescriptions, and telemedicine sessions.

## Architecture

Frontend: Angular; Backend: Java Spring Boot APIs; Database: MySQL; Messaging: Kafka; Deployment: Azure Kubernetes Service.

## Major Components

Patient Service, Doctor Service, Appointment Service, Prescription Service, Notification Service.

## Data Flow

Patients authenticate, search doctors, schedule appointments, doctors update prescriptions, and notifications are delivered via SMS and email.

## External Integrations

Twilio SMS, Video Conferencing API, Identity Provider using OAuth2.

## Security Requirements

MFA for doctors, encryption at rest, HIPAA-compliant audit trails, API gateway authentication, data masking.

## Assumptions

5,000 daily appointments, high availability, centralized logging and monitoring.